

Osteoarthritis

Osteoarthritis (OA) is the most common non-inflammatory arthritis. Its high prevalence is high in the elderly, and the high rate of disability related to disease makes it as a leading cause of disability. The occurrence of osteoarthritis is on the rise due to faulty lifestyle. Obesity specially imparts significant contribution to the weight bearing joints. Commonly affected joints include the cervical and lumbosacral spine, hip, knee, and first metatarsal phalangeal joint (MTP). In the hands, the distal and proximal interphalangeal joints and the base of the thumb are often affected. Usually affected joints are the wrist, elbow, and ankle. The symptoms can interfere with work and normal daily activities. Unlike some other types of arthritis, it does not affect the internal organs. Among those over 60 years old, about 10% of males and 18% of females are affected with OA. In the affected individual, osteoarthritis is the cause of about 2% of years lived with disability.

Osteoarthritis at a glance

The formation of hard knobs at the middle finger joints (known as Bouchard's nodes) and at the farthest joints of the fingers (known as Heberden's nodes) are a common feature of osteoarthritis in the hands.

Age group : Commonly observed in elderly age group

Symptoms : Joint pain, stiffness, joint swelling, joint creptus and decreased range of motion

Usual onset : Over years

Causes : Previous joint injury, abnormal joint or limb development, inherited factors.

Risk factors : Overweight, legs of different lengths, job with high levels of joint stress.

Diagnostic method : Based on symptoms, supported by other testing.

Treatment : Exercise, efforts to decrease joint stress, support groups, pain medications, joint replacement.

Frequency : 237 million / 3.3% (2015)

Definition

Osteoarthritis is the most common type of joint disease, affecting many people in the world. It can be thought of as a degenerative disorder arising from the biochemical breakdown of articular (hyaline) cartilage in the synovial joints. However, the current view holds that osteoarthritis involves not only the articular cartilage but also the entire joint organ, including the subchondral bone and synovium. OA is joint failure, a disease in which all structures of the joint have undergone pathologic change, often in concert. The pathologic sine qua non of disease is hyaline articular cartilage loss, present in a focal and, initially, nonuniform manner. This is accompanied by increasing thickness and sclerosis of the subchondral bony plate, by outgrowth of osteophytes at the joint margin, by stretching of the articular capsule, by mild synovitis in many affected joints, and by weakness of muscles bridging the joint. In knees, meniscal degeneration is part of the disease. There are numerous pathways that lead to joint failure, but the initial step is often joint injury in the setting of a failure of protective mechanisms.

Aetiology of OA

The daily stresses applied to the joints, especially the weight-bearing joints (eg, ankle, knee, and hip), play an important role in the development of osteoarthritis. External forces accelerate the catabolic effects of the chondrocytes and further disrupt the cartilaginous matrix.

Risk factors for osteoarthritis include the following:

- Age

- Obesity
- Trauma
- Genetics (significant family history)
- Reduced levels of sex hormones
- Muscle weakness
- Repetitive use (ie, jobs requiring heavy labor and bending)
- Infection
- Crystal deposition
- Acromegaly
- Previous inflammatory arthritis (eg, burnt-out rheumatoid arthritis)
- Heritable metabolic causes (eg, alkaptonuria, hemochromatosis, and Wilson disease)
- Hemoglobinopathies (eg, sickle cell disease and thalassemia)
- Neuropathic disorders leading to a Charcot joint (eg, syringomyelia, tabes dorsalis, and diabetes)
- Underlying morphologic risk factors (eg, congenital hip dislocation and slipped femoral capital epiphysis)
- Disorders of bone (eg, Paget disease and avascular necrosis)
- Previous surgical procedures (eg, meniscectomy)

Advancing age

With advancing age come reductions in cartilage volume, proteoglycan content, cartilage vascularization, and cartilage perfusion.

Obesity

Obesity increases the mechanical stress in a weight-bearing joint. It has been strongly linked to osteoarthritis of the knees and, to a lesser extent, of the hips. Obesity is associated with increased levels (both systemic and intra-articular) of adipokines (cytokines derived from adipose tissue), which may promote chronic, low-grade inflammation in joints.

Other causes

Trauma or surgery (including surgical repair of traumatic injury) involving the articular cartilage,

ligaments, or menisci can lead to abnormal biomechanics in the joints and accelerate osteoarthritis.

Muscle dysfunction compromises the body's neuromuscular protective mechanisms, leading to increased joint motion and ultimately resulting in osteoarthritis. Valgus malalignment at the knee has been shown to increase the incidence and risk of radiographic progression of knee osteoarthritis involving the lateral compartment.

Genetics

A hereditary component, particularly in osteoarthritis presentations involving multiple joints, has long been recognized.

Patho-physiology

The pathology of OA provides evidence of the involvement of many joint structures in disease. Cartilage initially shows surface fibrillation and irregularity. As disease progresses, focal erosions develop there, and these eventually extend down to the subjacent bone. With further progression, cartilage erosion down to bone expands to involve a larger proportion of the joint surface, even though OA remains a focal disease with non-uniform loss of cartilage.

After an injury to cartilage, chondrocytes undergo mitosis and clustering. While the metabolic activity of these chondrocyte clusters is high, the net effect of this activity is to promote proteoglycan depletion in the matrix surrounding the chondrocytes. This is because the catabolic is greater than the synthetic activity. As disease develops, collagen matrix becomes damaged, the negative charges of proteoglycans get exposed, and cartilage swells from ionic attraction to water molecules. Because in damaged cartilage proteoglycans are no longer forced into close proximity, cartilage does not bounce back after loading as it did when healthy, and cartilage becomes vulnerable to further injury. Chondrocytes at the basal level of cartilage undergo apoptosis.

With loss of cartilage come alterations in subchondral bone. Stimulated by growth factors and cytokines, osteoclasts and osteoblasts in the

subchondral bony plate, just underneath cartilage, become activated. Bone formation produces a thickening and stiffness of the subchondral plate that occurs even before cartilage ulcerates. Trauma to bone during joint loading may be the primary factor driving this bone response, with healing from injury (including microcracks) producing stiffness. Small areas of osteonecrosis usually exist in joints with advanced disease. Bone death may also be caused by bone trauma with shearing of microvasculature, leading to a cutoff of vascular supply to some bone areas.

At the margin of the joint, near areas of cartilage loss, osteophytes form. These start as outgrowths of new cartilage and, with neurovascular invasion from the bone, this cartilage ossifies. Osteophytes are an important radiographic hallmark of OA. In malaligned joints, osteophytes grow larger on the side of the joint subject to most loading stress (e.g., in varus knees, osteophytes grow larger on the medial side).

The synovium produces lubricating fluids that minimize shear stress during motion. In healthy joints, the synovium consists of a single discontinuous layer filled with fat and containing two types of cells, macrophages and fibroblasts, but in OA, it can sometimes become oedematous and inflamed. There is a migration of macrophages from the periphery into the tissue, and cells lining the synovium proliferate. Enzymes secreted by the synovium digest cartilage matrix that has been sheared from the surface of the cartilage.

Additional pathologic changes occur in the capsule, which stretches, becomes oedematous, and can become fibrotic.

Classification of Arthritis

A number of classification systems are used for gradation of osteoarthritis, some important are given as below.

- WOMAC scale, taking into account pain, stiffness and functional limitation.
- Kellgren-Lawrence grading scale for osteoarthritis of the knee. It uses only projectional radiography features.

- Tinnis classification for osteoarthritis of the hip joint, also using only projectional radiography features.
- Knee injury and Osteoarthritis Outcome Score (KOOS) and Hip disability and Osteoarthritis Outcome Score (HOOS) surveys.

Osteoarthritis can be classified into either primary or secondary depending on whether or not there is an identifiable underlying cause. Both primary generalized nodal osteoarthritis and erosive osteoarthritis (EOA, also called inflammatory osteoarthritis) are sub-sets of primary osteoarthritis. EOA is a much less common, and more aggressive inflammatory form of osteoarthritis which often affects the distal inter-phalangeal joints of the hand and has characteristic articular erosive changes on x-ray

Osteoarthritis can be classified on the basis of joint involvement:

- Hand:
 - o Trapeziometacarpal osteoarthritis
- Wrist (wrist osteoarthritis)
- Vertebral column (spondylosis)
 - o Facet joint arthrosis
- Hip osteoarthritis
- Knee osteoarthritis

Clinical Features

- Joint pain from OA is activity-related. Pain comes on either during or just after joint use and then gradually resolves. Early in disease, pain is episodic, triggered often by a day or two of overactive use of a diseased joint, such as a person with knee OA taking a long run and noticing a few days of pain thereafter. As disease progresses, the pain becomes continuous and even begins to be bothersome at night. Stiffness of the affected joint may be prominent, but morning stiffness is usually brief (<30 min).
- In knees, buckling may occur, in part, due to weakness of muscles crossing the joint. Mechanical symptoms, such as buckling, catching, or locking, could also signify internal derangement, such as meniscal tears, and need

to be evaluated. In the knee, pain with activities requiring knee flexion, such as stair climbing and arising from a chair, often emanates from the patellofemoral compartment of the knee, which does not actively articulate until the knee is bent 35°. Bursitis occurs commonly around knees and hips.

Physical Examination

Physical examination findings in patients with osteoarthritis are mostly limited to the affected joints. Following are the most common physical finding noted in cases of OA.

- Tenderness is over the affected joint line
- Reduced range of motion
- Crepitus are frequently present.
- Malalignment with a bony enlargement may occur.
- Bland effusion may be present over involved joints.
- Limitation of joint motion or muscle atrophy around a more severely affected joint may occur.
- Heberden nodes, which represent palpable osteophytes in the DIP joints, are more characteristic in women than in men.
- Inflammatory changes are mostly absent or at least not pronounced.

Progression of Osteoarthritis

The etiopathogenesis of osteoarthritis has been divided into 3 stages as follows:

- Stage 1 – Proteolytic breakdown of the cartilage matrix occurs
- Stage 2 – Fibrillation and erosion of the cartilage surface develop, with subsequent release of proteoglycan and collagen fragments into the synovial fluid
- Stage 3 – Breakdown products of cartilage induce a chronic inflammatory response in the synovium, which in turn contributes to further cartilage breakdown.

Ancillary Investigations

- Osteoarthritis is typically diagnosed on the basis of clinical and radiographic evidence.

No specific laboratory abnormalities are associated with osteoarthritis.

- Levels of acute-phase reactants are typically within the reference range in patients with osteoarthritis. The erythrocyte sedimentation rate (ESR) is not usually elevated, though it may be slightly so in cases of erosive inflammatory arthritis. The synovial fluid analysis usually shows a white blood cell (WBC) count below 2000/L, with a mononuclear predominance.

Osteoporosis

Osteoporosis is a silent disease that causes no symptoms until a fracture occurs. It is a major public health concern, with about 44 million American men and women, or 55% of the population age 50 and over, having osteoporosis or low bone density that can lead to fractures. About 80% of osteoporosis occurs in women and 20% in men. The prevalence is increasing, with over 61 million expected to have osteoporosis or low bone density by 2020. About 30% of Caucasian women age 50 and over have osteoporosis, when defined as a T-score of less than -2.5 at the spine, hip, or mid-forearm. In men age 50 and older, the prevalence of osteoporosis is about 19%. The Dubbo study found that at the age 60 there was a residual lifetime fracture risk of 56% for women and 29% for men, assuming average life expectancy. A woman's risk of hip fracture is equal to her combined risk of breast, uterine, and ovarian cancer. Fractures of the spine and hip are associated with an increased risk of chronic pain, deformity, depression, disability, and death. About 50% of those with a hip fracture will be permanently unable to walk without assistance and 25% will require long-term care. The mortality rate five years after a fracture of the hip or a clinical vertebral fracture is about 20% greater than expected, with men having higher mortality rates than women.

Risk factors

Risk factors for osteoporotic fracture can be split between non-modifiable and modifiable. In addition, osteoporosis is a recognized complication

of specific diseases and disorders. Medication use is theoretically modifiable, although in many cases, the use of medication that increases osteoporosis risk may be unavoidable. Caffeine is not a risk factor for osteoporosis. It is more common in females than males.

Non-modifiable risk factors

Bone density peaks at about 30 years of age. Women lose bone mass more rapidly than men.

- The most important risk factors for osteoporosis are advanced age (in both men and women) and female sex; oestrogen deficiency following menopause or surgical removal of the ovaries is correlated with a rapid reduction in bone mineral density, while in men, a decrease in testosterone levels has a comparable (but less pronounced) effect.
- **Ethnicity:** While osteoporosis occurs in people from all ethnic groups, European or Asian ancestry predisposes for osteoporosis.
- **Heredity:** Those with a family history of fracture or osteoporosis are at an increased risk; the heritability of the fracture, as well as low bone mineral density, is relatively high, ranging from 25 to 80%. At least 30 genes are associated with the development of osteoporosis.
- Those who have already had a fracture are at least twice as likely to have another fracture compared to someone of the same age and sex.
- **Build:** A small stature is also a non-modifiable risk factor associated with the development of osteoporosis.

Modifiable Factors

- **Excessive alcohol:** Although small amounts of alcohol are probably beneficial (bone density increases with increasing alcohol intake), chronic heavy drinking of probably increases fracture risk despite any beneficial effects on bone density.
- **Vitamin D deficiency:** Low circulating Vitamin D is common among the elderly worldwide. Mild vitamin D insufficiency is associated with increased parathyroid hormone

(PTH) production. PTH increases bone resorption, leading to bone loss. A positive association exists between serum 1,25-dihydroxycholecalciferol levels and bone mineral density, while PTH is negatively associated with bone mineral density.

- **Tobacco smoking:** Tobacco smoking has been assumed to inhibit the activity of osteoblasts, and is an independent risk factor for osteoporosis. Smoking also results in increased breakdown of exogenous estrogen, lower body weight and earlier menopause, all of which contribute to lower bone mineral density.
- **Malnutrition:** Nutrition has an important and complex role in maintenance of good bone. Identified risk factors include low dietary calcium and/or phosphorus, magnesium, zinc, boron, iron, fluoride, copper, vitamins A, K, E and C (and D where skin exposure to sunlight provides an inadequate supply). Excess sodium is a risk factor. High blood acidity may be diet-related, and is a known antagonist of bone. Some have identified low protein intake as associated with lower peak bone mass during adolescence and lower bone mineral density in elderly populations. Conversely, some have identified low protein intake as a positive factor, protein is among the causes of dietary acidity. Imbalance of omega-6 to omega-3 polyunsaturated fats is yet another identified risk factor.
- **High dietary protein from animal sources:** A diet high in plant protein may be optimal for bone health, as higher protein diets tend to increase absorption of calcium from the diet and are associated with higher bone density. Indeed, it has recently been argued that low protein diets cause poor bone health.
- **Underweight/inactive:** Bone remodelling occurs in response to physical stress, so physical inactivity can lead to significant bone loss. Weight bearing exercise can increase peak bone mass achieved in adolescence, and a highly significant correlation between bone strength and muscle strength has been

determined. The incidence of osteoporosis is lower in overweight people.

- **Endurance training:** In female endurance athletes, large volumes of training can lead to decreased bone density and an increased risk of osteoporosis. However, for male athletes, the situation is less clear, and although some studies have reported low bone density in elite male endurance athletes, others have instead seen increased leg bone density.
- **Heavy metals:** A strong association between cadmium and lead with bone disease has been established. Low-level exposure to cadmium is associated with an increased loss of bone mineral density readily in both genders, leading to pain and increased risk of fractures, especially in the elderly and in females. Higher cadmium exposure results in osteomalacia (softening of the bone).
- **Soft drinks:** Recent studies reveal that soft drinks may increase risk of osteoporosis, at least in women. Others suggest soft drinks may displace calcium-containing drinks from the diet rather than directly causing osteoporosis.
- Proton pump inhibitors (such as lansoprazole, esomeprazole, or omeprazole) that decrease stomach acid, are a risk for bone fractures if taken for two or more years, due to decreased absorption of calcium in the stomach.

Medical disorders may lead to Osteoporosis

The body regulates calcium homeostasis with two pathways; one is signalled to turn on when blood calcium levels drop below normal and one is the pathway that is signalled to turn on when blood calcium levels are elevated.

Many diseases and disorders have been associated with osteoporosis. Some important one is given below.

- In general, immobilization causes bone loss.
- Hypogonadal states can cause secondary osteoporosis.
- Endocrine disorders that can induce bone loss include Cushing's syndrome, hyperparathyroidism, hyperthyroidism, hypothyroidism, diabetes mellitus type 1 and 2, acromegaly,

and adrenal insufficiency.

- Malnutrition, parenteral nutrition and malabsorption can lead to osteoporosis. People with lactose intolerance or milk allergy may develop osteoporosis due to restrictions of calcium-containing foods. Individuals with bulimia can also develop osteoporosis. Those with an otherwise adequate calcium intake can develop osteoporosis due to the inability to absorb calcium and/or vitamin D. Other micronutrients such as vitamin K or vitamin B₁₂ deficiency may also contribute.
- People with rheumatologic disorders such as rheumatoid arthritis, ankylosing spondylitis, systemic lupus erythematosus and polyarticular juvenile idiopathic arthritis are at increased risk of osteoporosis.
- Chronic kidney disease can lead to renal osteodystrophy.
- Hematologic disorders linked to osteoporosis are multiple myeloma and other monoclonal gammopathies, lymphoma, leukemia, mastocytosis, hemophilia, sickle-cell disease and thalassemia.
- Several inherited disorders have been linked to osteoporosis. These include osteogenesis imperfecta, Marfan syndrome, hemochromatosis, etc.
- People with scoliosis of unknown cause also have a higher risk of osteoporosis. Bone loss can be a feature of complex regional pain syndrome. It is also more frequent in people with Parkinson's disease and chronic obstructive pulmonary disease.
- People with Parkinson's disease have a higher risk of broken bones. This is related to poor balance and poor bone density. In Parkinson's disease there may be a link between the loss of dopaminergic neurons and altered calcium metabolism.

Medication may lead to Osteoporosis

Certain medications have been associated with an increase in osteoporosis risk, which are described as below.

- Steroid-induced osteoporosis (SIOP) arises due

to use of glucocorticoids – analogous to Cushing's syndrome and involving mainly the axial skeleton. The synthetic glucocorticoid prescription drug prednisone is a main candidate after prolonged intake.

- Barbiturates, phenytoin and some other enzyme-inducing antiepileptics – these probably accelerate the metabolism of vitamin D.
- L-Thyroxine over-replacement may contribute to osteoporosis, in a similar fashion as thyrotoxicosis does. This can be relevant in subclinical hypothyroidism.
- Several drugs induce hypogonadism, for example aromatase inhibitors used in breast cancer, methotrexate and other antimetabolite drugs, depot progesterone and gonadotropin-releasing hormone agonists.
- Anticoagulants – long-term use of heparin is associated with a decrease in bone density, and warfarin (and related coumarins) have been linked with an increased risk in osteoporotic fracture in long-term use.
- Proton pump inhibitors – these drugs inhibit the production of stomach acid; this is thought to interfere with calcium absorption. Chronic phosphate binding may also occur with aluminium-containing antacids.
- Thiazolidinediones (used for diabetes) – rosiglitazone and possibly pioglitazone, inhibitors of PPAR γ , have been linked with an increased risk of osteoporosis and fracture.
- Chronic lithium therapy has been associated with osteoporosis.

Signs and symptoms

Illustration depicting normal standing posture and osteoporosis

Osteoporosis itself has no symptoms; its main consequence is the increased risk of bone fractures. Osteoporotic fractures occur in situations where healthy people would not normally break a bone; they are therefore regarded as fragility fractures. Typical fragility fractures occur in the vertebral column, rib, hip and wrist.

1. Fractures

Fractures are a common symptom of osteoporosis and can result in disability. Acute and chronic pain in the elderly is often attributed to fractures from osteoporosis and can lead to further disability and early mortality. These fractures may also be asymptomatic. The most common osteoporotic fractures are of the wrist, spine, shoulder and hip. The symptoms of a vertebral collapse are sudden back pain, often with radicular pain (shooting pain due to nerve root compression) and rarely with spinal cord compression or cauda equina syndrome. Multiple vertebral fractures lead to a stooped posture, loss of height, and chronic pain with resultant reduction in mobility.

Fractures of the long bones acutely impair mobility and may require surgery. Hip fracture, in particular, usually requires prompt surgery, as serious risks are associated with it, such as deep vein thrombosis and pulmonary embolism, and increased mortality.

Fracture risk calculators assess the risk of fracture based upon several criteria, including bone mineral density, age, smoking, alcohol usage, weight, and gender. The term “established osteoporosis” is used when a broken bone due to osteoporosis has occurred. Osteoporosis is a part of frailty syndrome.

Risk of fall

There is an increased risk of falls associated with aging. These falls can lead to skeletal damage at the wrist, spine, hip, knee, foot, and ankle. Part of the fall risk is because of impaired eyesight due to glaucoma, & macular degeneration, balance disorder, movement disorders, dementia, and sarcopenia (age-related loss of skeletal muscle). Collapse (transient loss of postural tone with or without loss of consciousness).

Pathogenesis

The underlying mechanism in all cases of osteoporosis is an imbalance between bone resorption and bone formation. In normal bone, matrix remodeling of bone is constant; up to 10% of all bone mass may be undergoing remodeling at

any point in time. The process takes place in bone multicellular units (BMUs) as first described by Frost & Thomas in 1963. Osteoclasts are assisted by transcription factor PU. to degrade the bone matrix, while osteoblasts rebuild the bone matrix. Low bone mass density can then occur when osteoclasts are degrading the bone matrix faster than the osteoblasts are rebuilding the bone.

The three main mechanisms by which osteoporosis develop are an inadequate peak bone mass (the skeleton develops insufficient mass and strength during growth), excessive bone resorption, and inadequate formation of new bone during remodeling. Interplay of these three mechanisms underlies the development of fragile bone tissue.

Hormonal factors strongly determine the rate of bone resorption; lack of estrogen (as a result of menopause) increases bone resorption, as well as decreasing the deposition of new bone that normally takes place in weight-bearing bones. The amount of estrogen needed to suppress this process is lower than that normally needed to stimulate the uterus and breast gland. The α -form of the estrogen receptor appears to be the most important in regulating bone turnover.

In addition to estrogen, calcium metabolism plays a significant role in bone turnover, and deficiency of calcium and vitamin D leads to impaired bone deposition; in addition, the parathyroid glands react to low calcium levels by secreting parathyroid hormone (parathormone, PTH), which increases bone resorption to ensure sufficient calcium in the blood. The role of calcitonin, a hormone generated by the thyroid that increases bone deposition, is less clear and probably not as significant as that of PTH.

Adhesive Capsulitis (Frozen Shoulder)

- Adhesive capsulitis and frozen shoulder syndrome (FSS) are two terms that have been used to describe a painful and stiff shoulder.
- The current consensus definition of a frozen shoulder by the American Shoulder and Elbow Surgeons is "a condition of uncertain etiology

characterized by significant restriction of both active and passive shoulder motion that occurs in the absence of a known intrinsic shoulder disorder."

- The American Academy of Orthopaedic Surgeons defines this condition as: "A condition of varying severity characterized by the gradual development of global limitation of active and passive shoulder motion where radiographic findings other than osteopenia are absent."
- The loss of passive range of motion (ROM) is a critical element in establishing the diagnosis of a true frozen shoulder. Although conditions such as subacromial bursitis, calcifying tendinitis, and partial rotator cuff tears can be associated with significant pain and loss of active ROM, passive ROM is preserved. Therefore, patients with those conditions should not be classified as having a frozen shoulder.
- Patients with primary frozen shoulder have no significant findings in the history, clinical examination, or radiographic evaluation to explain their motion loss and pain. Classically, symptoms of primary frozen shoulder have been divided into three phases: freezing (painful), frozen (stiffening), and thawing.

Stages of Frozen Shoulder Syndrome

Stage	Duration
Stage 1 Freezing (painful)	Up to 9 months
Stage 2 Frozen (stiffening)	4 months - 20 months
Stage 3 Thawing	5 months- 26 months
Contralateral shoulder	6 months-7 years after initial onset of symptoms

- Patients with diabetes and younger patients may be more likely to develop contralateral idiopathic adhesive capsulitis.

In contrast to patients with primary FSS, patients with secondary FSS describe an event that preceded the onset of shoulder symptoms, such as the following:

- Upper extremity trauma (eg, shoulder surgery, rotator cuff tear, proximal humerus fracture)
- Immobilization (eg, cardiothoracic surgery, neurosurgery)
- Metabolic/endocrine (eg, thyroid disease, diabetes mellitus, autoimmune disease, hyperlipidemia)
- Neurologic (eg, stroke, Parkinson disease)
- Cardiac disease (eg, ischemic heart disease, hypertension)
- Drugs (eg, protease inhibitors, antiretrovirals, immunizations, fluoroquinolones)
- Malignancy

Etiology

- Duplay was one of the first physicians to present the concept of periarticular tissue pathology rather than periarticular arthritis as the cause of frozen shoulder.
- Despite a lack of evidence linking frozen shoulder to a specific etiology, various triggers that may predispose patients to this problem appear to exist

A few reported etiologic agents include the following:

- Trauma
- Surgery (including but not limited to shoulder surgery)
- Inflammatory disease
- Diabetes
- Regional conditions
- Various shoulder maladies

Finally, frozen shoulder also has been associated with the following disorders:

- Cervical spine disease
- Parkinson disease
- Hyperthyroidism
- Hypothyroidism
- Ischemic heart disease

Clinical Presentation

Patients in the initial (freezing) stage of frozen shoulder syndrome (FSS) will describe the following:

- Insidious onset of vague, dull pain at the deltoid insertion
- Pain with shoulder movement
- Nagging pain at night, with sleep deprivation and the inability to sleep on the affected side
- Marked limitation of active and passive shoulder rotation, particularly external rotation
- The pain of FSS can be described as dull or aching. The pain leads to significant disability, affecting the activities of daily living, work, and leisure.
- During the second (frozen) stage, patients will complain of stiffness and severe loss of shoulder motion.
- The pain will be less pronounced. Patients will endorse the inability to reach over their head, to their side, and across their chest with the affected arm, or to scratch their back or put on a coat. The contralateral shoulder often becomes symptomatic years after FSS develops in the first shoulder.

Physical Examination

- The patient's posture should be observed while he or she is wearing a gown and sitting on a stool.
- On examination, the patient will have tenderness at the deltoid insertion and over the anterior capsule and posterior capsule with deep palpation.
- Compensatory scapulothoracic motion can create pain around the medial scapula. Forward flexion, abduction, and internal and external rotation of the shoulder should be assessed.
- Active and passive range of motion at the glenohumeral joint will be reduced, compared with the unaffected shoulder. Limitation of external rotation and abduction will be the most prominent findings.

Ancillary investigations & diagnosis

- Frozen shoulder syndrome (FSS) is a clinical diagnosis.
- Laboratory and imaging studies can be used to rule out other conditions and to confirm the

likelihood of the correct diagnosis. Accurate diagnosis is essential because the treatment for FSS differs from the treatment for other shoulder entities.

Laboratory Studies

- Laboratory studies rarely are required for the evaluation of adhesive capsulitis. However, if a predisposing medical condition that may be contributing to adhesive capsulitis is suggested, the following tests may be ordered:
- Complete blood cell count (CBC)
- Erythrocyte sedimentation rate (ESR)
- C-reactive protein (CRP)
- Blood glucose
- Hemoglobin A1c
- Thyroid-stimulating hormone (TSH)
- Free thyroxine index (FTI)
- Free thyroxine (FT4)

Imaging Studies

- Imaging studies are not indicated in the diagnosis of FSS.
- No imaging modality has been definitively shown to provide greater diagnostic value, due to the heterogeneity of techniques used, and the presence of potential confounding factors limits definitive conclusions from imaging study findings.
- Common findings in FSS are thickening of the coracohumeral and inferior glenohumeral ligaments.

Radiography

- Routine radiographs do not have a role in the diagnosis of FSS. However, plain films of the shoulder should be obtained in all cases to rule out any other pathologic process. These radiographs should include the following:
- Anteroposterior (AP) view of the glenohumeral joint in neutral rotation
- Supraspinatus outlet view
- Axillary lateral view (if possible)

Ultrasonography

- Ultrasound (regular and Doppler) for the diagnosis of FSS remains controversial.

- Overall, these studies have shown utility for measurement of acromioclavicular ligament thickness and presence of a hypoechoic region with increased vascularity in the rotator interval, with fibrovascular inflammatory tissue.

Arthrography

Arthrography, which involves intra-articular injection of diluted iodinated contrast medium, can aid in both the diagnosis and treatment of FSS.

Positron emission tomography/computed tomography

Duchstein et al suggest that 18-fluorodeoxyglucose positron emission tomography/computed tomography (^{18}F -FDG PET/CT) may have a role in diagnostically challenging cases.

Magnetic resonance imaging

- MRI is not initially indicated in cases of FSS. However, if the patient does not improve after a period of time (6 weeks to 3 months), then MRI is appropriate to rule out a possible rotator cuff tear or intra-articular pathology.

Histologic Findings

- Common arthroscopic findings in patients with adhesive capsulitis are the following:
- Proliferative synovitis
- Capsular and intra-articular subscapularis tendon thickening
- Fibrosis
- Chronic inflammatory cells

Lateral Epicondylitis (Tennis Elbow)

- Lateral epicondylitis (commonly referred to as "tennis elbow") is related to excessive wrist extension.
- It is the most common overuse syndrome.
- Patients typically report pain over the lateral elbow that worsens with activity and improves with rest.
- The symptoms tend to improve in 9-18 months; however, refractory cases may require surgery

Etiology

The causes of lateral epicondylitis include the following:

- Poor general conditioning leads to fatigue of the core and shoulder muscles, which puts an overemphasis on the extensor muscles of the forearm
- Improper training (eg, poor positioning when striking a tennis ball)
- Improper technique (eg, hitting a tennis ball late on the backhand)
- Poor or improper equipment (eg, a racquet that is strung too tightly)
- Scapular dyskinesis will lead to a compensatory increased load placed on the ipsilateral wrist extensors

Clinical Presentation

- The typical age of those affected is 40 to 50 years.
- Patients most typically report an insidious onset, but they will often relate a history of overuse without specific trauma.
- Symptom onset generally occurs 24-72 hours after repeated wrist extension activity.
- Delayed symptoms are probably due to microscopic tears in the tendon.
- The patient complains of pain over the lateral elbow that worsens with activity and improves with rest.
- The patient will also often describe aggravating conditions such as a backhand stroke in tennis or the overuse of a screwdriver.
- Pain may radiate down the posterior aspect of the forearm.
- Pain can vary from being mild (eg, with aggravating activities like tennis or the repeated use of a hand tool), or it can be such severe pain that simple activities like picking up and holding a coffee cup (ie, "coffee cup sign") will act as a trigger for the pain.

Physical Examination

The following may be noted on physical examination:

- Inspection: Very rarely does one notice swelling or ecchymosis.
- Palpation: Maximal tenderness is elicited 1-2

cm distal to the origin of the extensor carpi radialis brevis (ECRB) at the lateral epicondyle.

- Pain is increased with resisted wrist extension, with the wrist radially deviated and pronated and the elbow extended
- Increased pain is noted with resisted supination, gripping hand shaking.
- Always examine range of motion (ROM) of the shoulder, elbow, and wrist on the affected side.

Ancillary investigations -Diagnostic Studies

Laboratory studies

- Laboratory studies are typically not useful in the workup of lateral epicondylitis.

Imaging studies

- Consider plain film radiologic evaluation if the patient's symptoms persist despite adequate treatment or to evaluate for osteophytes, degenerative joint disease, or osteochondritis dissecans (OCD).
- Plain films may reveal osteophytes or calcifications along the lateral epicondyle.
- Consider magnetic resonance imaging (MRI), bone scanning, and/or computed tomography (CT) scanning to evaluate for OCD or stress fractures.
- Musculoskeletal ultrasonography is emerging as a useful modality to characterize areas of tendinosis, partial tearing, or calcifications and may assist with treatment options.

Other tests

A local anesthetic block may lead to symptom resolution and confirmation of the diagnosis.

Carpal Tunnel Syndrome

- Carpal tunnel syndrome (CTS) is a collection of characteristic symptoms and signs that occurs following compression of the median nerve within the carpal tunnel.
- It is characterised by numbness, paresthesias, and pain in the median nerve distribution. These symptoms may or may not be accompanied by objective changes in sensation

and strength of median-innervated structures in the hand.

Clinical presentation-Signs and symptoms

- Among the most common complaints, patients will reveal that their hands fall asleep or that things slip from their fingers without their noticing (loss of grip, dropping things); numbness and tingling also are commonly described.
- Such complaints should be localized to the palmar aspect of the first to the fourth fingers and the distal palm (ie, the sensory distribution of the median nerve at the wrist).
- The sensory symptoms above commonly are accompanied by an aching sensation over the ventral aspect of the wrist. This pain can radiate distally to the palm and fingers or, more commonly, extend proximally along the ventral forearm.
- Many patients with CTS also complain of a tight or swollen feeling in the hands and/or temperature changes (eg, hands being cold/hot all the time).
- Many patients also report sensitivity to changes in temperature (particularly cold) and a difference in skin color. In rare cases, there are complaints of changes in sweating.

Ancillary investigations and diagnosis

- Electrophysiologic studies, including electromyography (EMG) and nerve conduction studies (NCS), are the first-line investigations in suggested CTS.
- Abnormalities on electrophysiologic testing, in association with specific symptoms and signs, are considered the criterion standard for CTS diagnosis. In addition, other neurologic diagnoses can be excluded with these test results.
- Electrophysiologic testing also can provide an accurate assessment of how severe the damage to the nerve is, thereby directing management and providing objective criteria for the determination of prognosis.
- Ultrasonography potentially can identify

space-occupying lesions in and around the median nerve, confirm abnormalities in the median nerve (eg, increased cross-sectional area) that can be diagnostic of CTS, and help to guide steroid injections into the carpal tunnel.

- Magnetic resonance imaging (MRI) of the carpal tunnel is particularly useful preoperatively if a space-occupying lesion in the carpal tunnel is suggested.

Muscular Dystrophy

- Muscular dystrophy (MD) is a collective group of inherited noninflammatory but progressive muscle disorders without a central or peripheral nerve abnormality.
- The disease affects the muscles with definite fiber degeneration but without evidence of morphologic aberrations.
- Advances in molecular biology techniques illuminate the genetic basis underlying all types of MD: defects in the genetic code for dystrophin, a 427-kd skeletal muscle protein (Dp427).
- These defects result in the various manifestations commonly associated with MD, such as weakness and pseudohypertrophy.
- Dystrophin can also be found in cardiac smooth muscles and in the brain (accounting for the slight mental retardation associated with this disease).
- Minor variations notwithstanding, all types of MD have in common progressive muscle weakness that tends to occur in a proximal-to-distal direction, though there are some rare distal myopathies that cause predominantly distal weakness.
- The decreasing muscle strength in those who are affected may compromise the patient's ambulation potential and, eventually, cardiopulmonary function.
- Once wheelchair-bound, patients with MDs tend to develop worsening contractures, osteopenia (initially, followed by osteoporosis), and rapidly progressive scoliosis.

- The change in bone mineral density (BMD) can lead to fractures of the long bones or vertebrae, which in turn lead to more immobilization, disuse demineralization, and exacerbation of scoliosis.
 - The goal of orthopedic management is, therefore, to preserve or prolong patients' ambulatory status for as long as possible.
- Etiology

Classification of types of muscular dystrophy

- The etiology of MD is an abnormality in the genetic code for specific muscle proteins.
- They all are classified according to the clinical phenotype, the pathology, and the mode of inheritance.
- The inheritance pattern includes the sex-linked, autosomal recessive, and autosomal dominant MDs. Within each group of heritable MDs several disorders exist. These are characterized by the clinical presentation and pathology.

Heritable MDs include the following:

- Sex-linked MDs - Duchenne, Becker, Emery-Dreifuss
- Autosomal dominant MDs - Facioscapulohumeral, distal, ocular, oculopharyngeal
- Autosomal recessive MD Limb-girdle form

Genetic defects and dystrophin

In the X-linked forms of MD, such as the Duchenne and Becker dystrophies, the defect is located on the short arm of the X chromosome. Dystrophin is distributed not only in skeletal muscle but also in smooth and cardiac muscles and in the brain. The large size of the dystrophin gene explains the ease at which spontaneous new mutations can occur, as in Duchenne MD. The large size also allows mistakes in protein synthesis to occur at multiple sites.

Clinical presentation History and Physical Examination

Three major time points for patients with Duchenne MD are as follows :

- When they begin to walk
- When they lose their ability to ambulate

- When they die

Phases of Duchenne muscular dystrophy

- The child's motor milestones may be at the upper limits of normal, or they may be slightly delayed.
- Some of the delays may be caused by inherent muscle weakness, but a component may stem from brain involvement.
- Other early features include a gait abnormality, which classically is a waddling, wide-based gait with hyperlordosis of the lumbar spine and toe walking. The waddle is due to weakness in the gluteus maximus and gluteus medius muscles and the patient's inability to support a single-leg stance. The child leans the body toward the other side to balance the center of gravity, and the motion is repeated with each step.
- Hip extensor weakness also results in a forward tilt of the pelvis, which translates to a hyperlordosis of the spine to maintain posture.
- Gradually, the child is observed to have increasingly noticeable difficulty with step taking.
- The child then begins having problems getting up from the sitting or supine position, and he or she can rise to an upright stance only by manifesting the Gowers sign.
- The Gowers sign is a classic physical examination finding in MD and results from weakness in the child's proximal hip muscles.
- Although the Gowers sign is a classic finding in Duchenne MD, it is by no means pathognomonic; other types of MD and disorders with proximal weakness may also cause this sign.
- While still ambulatory, the child may have minimal deformities, including iliopsoas or tendo Achillis tightness.
- The second important phase in Duchenne MD is the loss of ambulation.
- With the child's loss of ambulation, there is usually a rapidly progressive course of muscle or tendon contractures and scoliosis.

- Duchenne MD is a terminal disease in which death usually occurs by the third decade of life (mostly from cardiopulmonary compromise), despite steroid treatment.
- The respiratory failure can easily occur from the underlying progressive nocturnal hypoventilation and hypoxia or from an acute cardiac insufficiency.

Additional clinical findings in Duchenne muscular dystrophy

- Other clinical findings in Duchenne MD include absent deep tendon reflexes in the upper extremities and patella (though the tendo Achillis reflex remains intact even in the later stages of this disease), pain in the calves with activity, pseudohypertrophy of the calf and macroglossia.
- Cardiopulmonary involvement is present from the beginning of the disease stages, but the findings are not so clinically obvious. Electrocardiographic (ECG) tracings show right ventricular strain, tall R waves, deep Q waves, and inverted T waves.

Other types of muscular dystrophy

- Becker MD is similar to Duchenne MD, but because patients have some measure of functioning dystrophin, the manifestations of Becker MD occur later and are milder. Patients tend to live past the fourth or fifth decades.
- Emery-Dreifuss MD is an uncommon sex-linked dystrophy that presents with early contractures and cardiomyopathy in affected patients; the typical presentation involves tendo Achillis contractures, elbow flexion contractures, neck extension contractures, tightness of the lumbar paravertebral muscles, and cardiac abnormalities.
- Death may occur in the fourth or fifth decade as a result of first-degree atrioventricular (AV) block, a condition that is usually not present at the initial presentation of this disease.
- None of the autosomal dominant conditions significantly affect longevity.

Ancillary investigations-Laboratory Studies

- A creatine phosphokinase (CPK) determination is the most specific test for muscular dystrophy (MD).
- Elevated CPK levels are indicative of muscle disease. Because the concentration of CPK is not significant in red blood cells, CPK levels are not affected by hemolysis.
- CPK is not affected by liver dysfunction, as are the other tested enzymes (eg, transaminases, aldolase, lactate dehydrogenase).
- High CPK levels represent leakage of the enzyme from the muscle cells only. This change is not exactly correlated with the severity of the disease.
- All MDs result in some CPK elevation during the active phase of the disease. The finding of three elevated levels obtained 1 month apart is diagnostic for MD. Early in the disease process, CPK levels are 50-300 times greater than normal levels, but the levels tend to decrease as the muscle mass decreases. The CPK level is highest in Duchenne MD, with less elevation noted in Becker MD.

Enzyme levels that may be elevated but can be altered by liver dysfunction include the following:

- Transaminase levels
- Lactate dehydrogenase levels
- Aldolase levels
- PCR can be used to detect more than 98% of existing deletions, and it can be performed within 24 hours.

Imaging Studies

- Ultrasonography (US) is a relatively noninvasive technique that is used for screening patients with MD; this modality is rapidly replacing electromyography (EMG) in centers that have appropriately trained staff.

Other Tests

- Electrocardiography (ECG) is expected to show a right ventricular strain, tall R waves, deep Q waves, and inverted T waves.
- Pulmonary function tests (PFTs), including an

analysis of arterial blood gases, and a hematologic workup are necessary as part of the preoperative workup.

- EMG usually demonstrates short-duration, polyphasic, motor-unit action potentials with decreased amplitudes. It should be kept in mind that this finding is common with all myopathic processes and does not specifically identify MDs.

Biopsy

- Until the advent of molecular biology techniques, muscle biopsy was the definitive test for diagnosing and confirming muscular disease.
- Histologic changes depend on the stage of disease and the muscle selected. The optimal site for biopsy is the vastus lateralis, accessed via a small lateral thigh incision.

Histologic Findings

- Histologic specimens from muscle biopsy samples obtained early in the development of MD show only variations in muscle fiber sizes with focal areas of degenerating or regenerating fibers. In later stages of MD, the changes are more obvious, with marked

variations in muscle fiber sizes, degeneration, and regeneration.

MCQs

1. आहारपरिणामान्तेभूयश्चलभतेवलम is clinical feature of
 - a. व्यपेताहिक्या (Vyapeta Hikk)
1. Carpal tunnel syndrome can be diagnosed by...
 - a. No Visual examination by a doctor is required only Nerve conduction velocity test (NCV)
 - b. Manual Muscle Testing (MMT)
 - c. Electrocardiography (ECG)
 - d. Visual examination by a doctor, Nerve conduction velocity test (NCV) & Electromyogram (EMG)
2. Which of these is not a form of muscular dystrophy?
 - a. Duchenne muscular dystrophy
 - b. Facioscapulohumeral dystrophy
 - c. Lunar dystrophy
 - d. Myotonic dystrophy
3. Nerve affected in Carpal tunnel syndrome
 - a. Median nerve
 - b. Radial nerve
 - c. Ulnar nerve
 - d. Musculocutaneous nerve

Answer

1.	d	2.	c	3.	a
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